

Proof. 1. **Degenerate cases.** If $x = 0$ then $\|x + y\| = \|y\| = \|0\| + \|y\|$; similarly if $y = 0$. Hence the inequality holds with equality. Henceforth assume $x \neq 0$ and $y \neq 0$.

2. **Expand $\|x + y\|^2$.** By definition of the norm and bilinearity of the inner product,

$$\begin{aligned}\|x + y\|^2 &= \langle x + y, x + y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle.\end{aligned}$$

Since $\langle y, x \rangle = \overline{\langle x, y \rangle}$,

$$\begin{aligned}\|x + y\|^2 &= \|x\|^2 + \|y\|^2 + \langle x, y \rangle + \overline{\langle x, y \rangle} \\ &= \|x\|^2 + \|y\|^2 + 2\Re\langle x, y \rangle.\end{aligned}\tag{1}$$

3. **Estimate the middle term.** For any complex number z ,

$$|\Re z| \leq |z|.\tag{2}$$

Indeed, $|z|^2 = (\Re z)^2 + (\Im z)^2 \geq (\Re z)^2$, whence $|\Re z| \leq |z|$. Applying (2) with $z = \langle x, y \rangle$ gives

$$\Re\langle x, y \rangle \leq |\langle x, y \rangle|.\tag{3}$$

Now invoke the Cauchy–Schwarz inequality:

$$|\langle x, y \rangle| \leq \|x\| \|y\|.\tag{4}$$

Combining (3) and (4) we obtain

$$\Re\langle x, y \rangle \leq \|x\| \|y\|.\tag{5}$$

4. **Upper bound for $\|x + y\|^2$.** Insert (5) into (1):

$$\begin{aligned}\|x + y\|^2 &\leq \|x\|^2 + 2\|x\| \|y\| + \|y\|^2 \\ &= (\|x\| + \|y\|)^2.\end{aligned}\tag{6}$$

All quantities involved are non-negative.

5. **Take square roots.** Since the square-root function is monotone on $[0, \infty)$,

$$\|x + y\| \leq \|x\| + \|y\|.$$

6. **Equality condition.** Equality in the chain (6) occurs precisely when each inequality used becomes an equality.

- From (5): equality in $\Re\langle x, y \rangle \leq |\langle x, y \rangle|$ forces $\langle x, y \rangle$ to be a non-negative real number (i.e. $\Im\langle x, y \rangle = 0$ and $\Re\langle x, y \rangle \geq 0$).

- Equality in Cauchy–Schwarz (4) holds iff x and y are linearly dependent, i.e. $y = \lambda x$ for some $\lambda \in \mathbb{C}$.
- Substituting $y = \lambda x$ gives $\langle x, y \rangle = \lambda \langle x, x \rangle$; this is real and non-negative exactly when λ is a non-negative real scalar.

Thus equality $\|x + y\| = \|x\| + \|y\|$ holds iff one of the following is true:

- (a) $x = 0$ or $y = 0$ (the degenerate case);
- (b) There exists $\lambda \in \mathbb{R}$ with $\lambda \geq 0$ such that $y = \lambda x$ (the vectors are positively colinear).

These are precisely the cases announced in the statement of the triangle inequality.

□